glmCH2_extra

Create dataframe where each row represents number of queens and workers seen during each survey event

Load dataframes

```r
#load packages
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.0     v stringr   1.5.1
## v ggplot2   3.5.2     v tibble    3.3.0
## v lubridate 1.9.4     v tidyr     1.3.1
## v purrr     1.0.4
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
library(lubridate) #package used to generate julian date from calendar date
library(ggplot2)


#read in data
bee <- read.csv("06_cleandata/bee_clean.csv")
floral <- read.csv("06_cleandata/floral_clean.csv")
rodent <- read.csv("06_cleandata/rodent_cleanv2.csv")
veg <- read.csv("06_cleandata/veg_clean.csv")
site_info <- read.csv("06_cleandata/site_info_clean.csv")
surveys <- read.csv("06_cleandata/surveys.csv")
```

Fix dataframe structure

```r
#bee
str(bee)
```

```
## 'data.frame':    520 obs. of  17 variables:
##  $ date            : chr  "2024-04-13" "2024-04-13" "2024-04-13" "2024-04-13" ...
##  $ observers       : chr  "RG" "RG" "RG" "RG" ...
##  $ site_id         : chr  "chong_96" "chong_96" "chong_96" "chong_96" ...
```

```
##  $ round            : chr  "2024_r1" "2024_r1" "2024_r1" "2024_r1" ...
##  $ plot_id          : chr  "p1" "p2" "d" "d" ...
##  $ transect_id      : chr  "full-plot" "full-plot" "full-plot" "full-plot" ...
##  $ transect_section: chr  "net-floral" "net-floral" "net-floral" "net-floral" ...
##  $ sample_code      : chr  "chong_96_NS_p1_full-plot_net-floral" "chong_96_NS_p2_full-plot_net-floral
##  $ behaviour        : chr  "flying" "flying" "flying" "nest_searching" ...
##  $ caste            : chr  "q" "q" "q" "q" ...
##  $ flower_code      : chr  "" "" "" "" ...
##  $ flower_species   : chr  "" "" "" "" ...
##  $ bee_code         : chr  "" "" "boim" "bovo" ...
##  $ bee_species      : chr  "" "" "bombus_impatiens" "bombus_vosnesenskii" ...
##  $ caught           : chr  "n" "n" "y" "n" ...
##  $ photo_series     : chr  "" "" "" "" ...
##  $ notes            : chr  "could not ID" "could not ID" "" "" ...
```

```r
bee$date <- ymd(bee$date)

#floral
str(floral)
```

```
## 'data.frame':    786 obs. of  12 variables:
##  $ date             : chr  "2024-04-13" "2024-04-16" "2024-04-16" "2024-04-18" ...
##  $ observers        : chr  "RG" "RG" "CN" "RG" ...
##  $ site_id          : chr  "chong_96" "bates_57b" "bates_57b" "bks_96" ...
##  $ round            : chr  "2024_r1" "2024_r1" "2024_r1" "2024_r1" ...
##  $ plot_id          : chr  "p2" "p1" "p2" "p1" ...
##  $ transect_id      : chr  "full-plot" "full-plot" "full-plot" "full-plot" ...
##  $ transect_section: chr  "net-floral" "net-floral" "net-floral" "net-floral" ...
##  $ sample_code      : chr  "chong_96_NS_p2_full-plot_net-floral" "bates_57b_NS_p1_full-plot_net-floral
##  $ flower_code      : chr  "taof" "taof" "taof" "vaco" ...
##  $ flower_species   : chr  "taraxacum_officinale" "taraxacum_officinale" "taraxacum_officinale" "vacci
##  $ abundance        : int  2 4 4 4 1 4 3 4 4 2 ...
##  $ notes            : chr  "" "" "" "" ...
```

```r
floral$date <- ymd(floral$date)

#rodent
str(rodent)
```

```
## 'data.frame':    1432 obs. of  25 variables:
##  $ date_day                  : chr  "2024-04-17" "2024-04-17" "2024-04-17" "2024-04-17" ...
##  $ observers_day             : chr  "RG, AD" "RG, AD" "RG, AD" "RG, AD" ...
##  $ date_night                : chr  "" "" "" "" ...
##  $ observers_night           : chr  "" "" "" "" ...
##  $ site_id                   : chr  "gwb_neaves" "gwb_neaves" "gwb_neaves" "gwb_neaves" ...
##  $ round                     : chr  "2024_r1" "2024_r1" "2024_r1" "2024_r1" ...
##  $ plot_id                   : chr  "d" "d" "d" "d" ...
##  $ transect_id               : chr  "t2" "t2" "t2" "t2" ...
##  $ transect_section          : chr  "rod-s1" "rod-s1" "rod-s1" "rod-s1" ...
##  $ burrow_code               : chr  "gwb_neaves__d_t2_rod-s1" "gwb_neaves__d _t2_rod-s1" "gwb_neaves
##  $ location_m                : num  0 0.2 0.4 0.5 5 7.5 6 5 20 20.5 ...
##  $ location_side             : chr  "" "" "" "" ...
##  $ row_alley                 : chr  "" "" "" "" ...
```

```
##  $ diameter_cm               : num  2.5 5 2.5 5 2.5 2.5 2.5 2.5 5 5 ...
##  $ species_guess             : chr  "" "" "" "" ...
##  $ substrate                 : chr  "" "" "" "" ...
##  $ notes_hole                : chr  "rocky" "rocky" "rocky" "rocky" ...
##  $ powder_active_field       : chr  "" "" "" "" ...
##  $ powder_species_guess_field: chr  "" "" "" "" ...
##  $ notes_powder              : chr  "" "" "" "" ...
##  $ photo_number              : chr  "" "" "" "" ...
##  $ photo_location            : chr  "" "" "" "" ...
##  $ powder_active_FINAL       : chr  "" "" "" "" ...
##  $ powder_species_guess_FINAL: chr  "" "" "" "" ...
##  $ notes_extra_FINAL         : chr  "" "" "" "" ...
```

```
rodent$date_day <- ymd(rodent$date_day)

#veg
str(veg)
```

```
## 'data.frame':    2802 obs. of  128 variables:
##  $ date        : chr  "2024-04-13" "2024-04-13" "2024-04-13" "2024-04-13" ...
##  $ observers   : chr  "RG" "RG" "RG" "RG" ...
##  $ site_id     : chr  "chong_96" "chong_96" "chong_96" "chong_96" ...
##  $ round       : chr  "2024_r1" "2024_r1" "2024_r1" "2024_r1" ...
##  $ plot_id     : chr  "p2" "p2" "p2" "p2" ...
##  $ transect_id : chr  "t1" "t2" "t1" "t2" ...
##  $ quadrat     : chr  "veg-q1m" "veg-q1e" "veg-q3m" "veg-q1m" ...
##  $ sample_code : chr  "chong_96_NS_p2_t1_veg-q1m" "chong_96_NS_p2_t2_veg-q1e" "chong_96_NS_p2_t1_veg
##  $ bare        : num  23 25 35 35 35 35 35 45 45 50 ...
##  $ dist        : int  0 0 0 0 0 0 0 0 0 0 ...
##  $ grass       : num  70 2 60 45 60 10 0 10 0.5 20 ...
##  $ forb        : num  5 4 0 0 0 0 0 0 0 0 ...
##  $ grass_height: num  10 7 16 14 14 8 NA 7 3 10 ...
##  $ forb_height : num  6 4 NA NA NA NA NA NA NA NA ...
##  $ mulch       : int  0 50 0 0 0 45 45 35 50 0 ...
##  $ dry_root    : num  2 20 5 20 5 10 20 10 5 30 ...
##  $ flood       : num  0 0 0 0 0 0 0 0 0 0 ...
##  $ dry.unroot  : int  NA NA NA NA NA NA NA NA NA NA ...
##  $ bryophyte   : num  0 0 0 0 0 0 0 0 0 0 ...
##  $ live_mow    : int  NA NA NA NA NA NA NA NA NA NA ...
##  $ rock        : int  0 0 0 0 0 0 0 0 0 0 ...
##  $ gravel      : int  0 0 0 0 0 0 0 0 0 0 ...
##  $ garbage     : int  0 0 0 0 0 0 0 0 0 0 ...
##  $ vaco_perc   : num  0 0 0 0 0 0 0 0 0 0 ...
##  $ vaco_num    : num  0 0 0 0 0 0 0 0 0 0 ...
##  $ anco_perc   : num  0 0 0 0 0 0 0 0 0 0 ...
##  $ anco_num    : int  0 0 0 0 0 0 0 0 0 0 ...
##  $ arth_perc   : num  0 0 0 0 0 0 0 0 0 0 ...
##  $ arth_num    : int  0 0 0 0 0 0 0 0 0 0 ...
##  $ brra_perc   : int  0 0 0 0 0 0 0 0 0 0 ...
##  $ brra_num    : int  0 0 0 0 0 0 0 0 0 0 ...
##  $ cabu_perc   : num  0 0 0 0 0 0 0 0 0 0 ...
##  $ cabu_num    : int  0 0 0 0 0 0 0 0 0 0 ...
##  $ cahi_perc   : num  0 0 0 0 0 0 0 0 0 0 ...
##  $ cahi_num    : int  0 0 0 0 0 0 0 0 0 0 ...
```

```
##  $ cape_perc   : num  0 0 0 0 0 0 0 0 0 0 ...
##  $ cape_num    : int  0 0 0 0 0 0 0 0 0 0 ...
##  $ cegl_perc   : num  0 0 0 0 0 0 0 0 0 0 ...
##  $ cegl_num    : int  0 0 0 0 0 0 0 0 0 0 ...
##  $ clpa_perc   : num  0 0 0 0 0 0 0 0 0 0 ...
##  $ clpa_num    : int  0 0 0 0 0 0 0 0 0 0 ...
##  $ crdo_perc   : int  0 0 0 0 0 0 0 0 0 0 ...
##  $ crdo_num    : int  0 0 0 0 0 0 0 0 0 0 ...
##  $ drve_perc   : num  0 0 0 0 0 0 0 0 0 0 ...
##  $ drve_num    : int  0 0 0 0 0 0 0 0 0 0 ...
##  $ epci_perc   : num  0 0 0 0 0 0 0 0 0 0 ...
##  $ epci_num    : int  0 0 0 0 0 0 0 0 0 0 ...
##  $ gedi_perc   : num  0 0 0 0 0 0 0 0 0 0 ...
##  $ gedi_num    : int  0 0 0 0 0 0 0 0 0 0 ...
##  $ hypo_perc   : num  0 0 0 0 0 0 0 0 0 0 ...
##  $ hypo_num    : int  0 0 0 0 0 0 0 0 0 0 ...
##  $ lapu_perc   : num  0 0 0 0 0 0 0 0 0 0 ...
##  $ lapu_num    : int  0 0 0 0 0 0 0 0 0 0 ...
##  $ lope_perc   : num  0 0 0 0 0 0 0 0 0 0 ...
##  $ lope_num    : int  0 0 0 0 0 0 0 0 0 0 ...
##  $ madi_perc   : num  0 0 0 0 0 0 0 0 0 0 ...
##  $ madi_num    : int  0 0 0 0 0 0 0 0 0 0 ...
##  $ moli_perc   : num  0 0 0 0 0 0 0 0 0 0 ...
##  $ moli_num    : int  0 0 0 0 0 0 0 0 0 0 ...
##  $ mydi_perc   : num  0 0 0 0 0 0 0 0 0 0 ...
##  $ mydi_num    : int  0 0 0 0 0 0 0 0 0 0 ...
##  $ myst_perc   : num  0 0 0 0 0 0 0 0 0 0 ...
##  $ myst_num    : int  0 0 0 0 0 0 0 0 0 0 ...
##  $ pela_perc   : num  0 0 0 0 0 0 0 0 0 0 ...
##  $ pela_num    : int  0 0 0 0 0 0 0 0 0 0 ...
##  $ pema_perc   : num  0 0 0 0 0 0 0 0 0 0 ...
##  $ pema_num    : int  0 0 0 0 0 0 0 0 0 0 ...
##  $ plla_perc   : num  0 0 0 0 0 0 0 0 0 0 ...
##  $ plla_num    : int  0 0 0 0 0 0 0 0 0 0 ...
##  $ poav_perc   : num  0 0 0 0 0 0 0 0 0 0 ...
##  $ poav_num    : int  0 0 0 0 0 0 0 0 0 0 ...
##  $ prkn_perc   : int  0 0 0 0 0 0 0 0 0 0 ...
##  $ prkn_num    : int  0 0 0 0 0 0 0 0 0 0 ...
##  $ raac_perc   : num  0 0 0 0 0 0 0 0 0 0 ...
##  $ raac_num    : int  0 0 0 0 0 0 0 0 0 0 ...
##  $ rara_perc   : num  0 0 0 0 0 0 0 0 0 0 ...
##  $ rara_num    : int  0 0 0 0 0 0 0 0 0 0 ...
##  $ rasa_perc   : int  0 0 0 0 0 0 0 0 0 0 ...
##  $ rasa_num    : int  0 0 0 0 0 0 0 0 0 0 ...
##  $ rare_perc   : num  0 0 0 0 0 0 0 0 0 0 ...
##  $ rare_num    : int  0 0 0 0 0 0 0 0 0 0 ...
##  $ rasc_perc   : num  0 0 0 0 0 0 0 0 0 0 ...
##  $ rasc_num    : int  0 0 0 0 0 0 0 0 0 0 ...
##  $ ropa_perc   : num  0 0 0 0 0 0 0 0 0 0 ...
##  $ ropa_num    : int  0 0 0 0 0 0 0 0 0 0 ...
##  $ ruac_perc   : int  0 0 0 0 0 0 0 0 0 0 ...
##  $ ruac_num    : int  0 0 0 0 0 0 0 0 0 0 ...
##  $ rula_perc   : num  0 0 0 0 0 0 0 0 0 0 ...
##  $ rula_num    : int  0 0 0 0 0 0 0 0 0 0 ...
```

```
##  $ ruur_perc   : int  0 0 0 0 0 0 0 0 0 0 ...
##  $ ruur_num    : int  0 0 0 0 0 0 0 0 0 0 ...
##  $ ruul_perc   : num  0 0 0 0 0 0 0 0 0 0 ...
##  $ ruul_num    : int  0 0 0 0 0 0 0 0 0 0 ...
##  $ sara_perc   : int  0 0 0 0 0 0 0 0 0 0 ...
##  $ sara_num    : int  0 0 0 0 0 0 0 0 0 0 ...
##  $ scau_perc   : num  0 0 0 0 0 0 0 0 0 0 ...
##  $ scau_num    : int  0 0 0 0 0 0 0 0 0 0 ...
##  $ stme_perc   : num  0 0 0 0 0 0 0 0 0 0 ...
##  $ stme_num    : int  0 0 0 0 0 0 0 0 0 0 ...
##   [list output truncated]
```

```r
veg$date <- ymd(veg$date)

#site_info
str(site_info)
```

```
## 'data.frame':    12 obs. of  8 variables:
##  $ site_id   : chr  "bates_57b" "bks_56" "chong_68" "bks_68" ...
##  $ area      : chr  "delta_1" "delta_1" "delta_2" "delta_2" ...
##  $ grower    : chr  "George Bates" "Tejinder" "Nancy Chong" "Tejinder" ...
##  $ company   : chr  "Bate's Farm" "BKS Farms" "Chong Farms" "BKS Farms" ...
##  $ address   : chr  "Deltaport Way and 53 St" "2601 56th St" "6638 68th Street, Delta" "6570 68th st
##  $ cover_type: chr  "grass" "bare" "grass" "bare" ...
##  $ x         : num  -123 -123 -123 -123 -123 ...
##  $ y         : num  49.1 49.1 49.1 49.1 49.1 ...
```

```r
#surveys
str(surveys)
```

```
## 'data.frame':    210 obs. of  3 variables:
##  $ site : chr  "bates_57b" "bates_57b" "bates_57b" "bates_57b" ...
##  $ round: chr  "2024_r1" "2024_r1" "2024_r1" "2024_r2" ...
##  $ plots: chr  "p1" "p2" "d" "p1" ...
```

```r
surveys <- surveys %>%
  rename(site_id = site)
surveys <- surveys %>%
  rename(plot_id = plots)
```

Add date of bee survey to the associated row in the survey data, based on site, plot and round.

```r
# Extract relevant bee dates first
bee_dates <- bee %>%
  dplyr::select(site_id, plot_id, round, date) %>%
  dplyr::rename(bee_date = date)

#Make bee_dates file only contain one row for each site/plot/round/date combination
bee_dates_unique <- bee_dates %>%
```

```
  group_by(site_id, round) %>%
  summarise(bee_date = first(bee_date), .groups = "drop") #choose first date since all dates within a s

#Join bee_date onto survey data

surveys <- surveys %>% left_join(bee_dates_unique, by = c("site_id", "round"))

#Need to manually add in the dates for 2 surveys where I didn't find any bees (surjit_168 during 2025_r
missing_dates <- tibble::tribble(
  ~site_id, ~round, ~bee_date,
  "surjit_168", "2025_r2", as.Date("2025-05-13"),
  "bks_neaves", "2025_r1", as.Date("2025-04-14"),
  "bks_56", "2024_r3", as.Date("2025-06-21"),
  "bks_56", "2025_r3", as.Date("2025-06-18"),
  "bks_neaves", "2025_r3", as.Date("2025-06-24"),
  "surjit_168", "2025_r3", as.Date("2025-06-25"),
  "sandhu_ladnertrunk", "2025_r3", as.Date("2025-06-23") )

surveys <- surveys %>% rows_patch(missing_dates, by = c("site_id", "round"))

#add column for julian date
surveys$julian.date <- yday(surveys$bee_date)
```

Add area and cover type to survey dataframe based on site_id

```
surveys <- surveys %>%
  left_join(site_info %>%
              dplyr::select(site_id, area, cover_type), by = c("site_id"))
```

Adding all the desired variables summarized at plot level as columns to the surveys dataframe

The desired variables are bee abundance for each caste of bees (queen and worker)

```
# 1. Summarize your actual bee observations
bee_counts <- bee %>%
  filter(caste %in% c("q", "w")) %>%
  group_by(site_id, round, plot_id, caste) %>%
  summarise(n_bees = n(), .groups = "drop")

# 2. Use your existing 'surveys' dataframe as the base
# This ensures you ONLY have rows for actual survey events
surveys <- surveys %>%
  # Create a slot for both 'q' and 'w' for every real survey row
  crossing(caste = c("q", "w")) %>%
  # Join the counts
  left_join(bee_counts, by = c("site_id", "round", "plot_id", "caste")) %>%
  # Fill the zeros for the surveys where that caste wasn't seen
  mutate(n_bees = replace_na(n_bees, 0))
```

Run model to see the effect of date, caste, and plot type on bee abundance.

1. Load packages and clean dataframe

```
#Load packages
library(DHARMa) #diagnostic tools for mixed regression models
```

```
## This is DHARMa 0.4.7. For overview type '?DHARMa'. For recent changes, type news(package = 'DHARMa')
```

```
library(glmmTMB) #fits generalized linear mixed models, including zero-inflated and overdispersed model

#clean surveys dataframe and scale continuous predictors

#change character columns to factors
surveys <- surveys %>%
   mutate(
     site_id = as.factor(site_id),
     round = as.factor(round),
     plot_id = as.factor(plot_id),
     bee_date = as.Date(bee_date),
     area = as.factor(area),
     cover_type = as.factor(cover_type))

#add plot type column and make it a factor
surveys <- surveys %>%
  mutate(plot_type = if_else(plot_id == "d", "ditch", "field"))
surveys <- surveys %>%
   mutate(
     plot_type = as.factor(plot_type))

#scale continuous predictors
surveys$julian.date_scaled <- as.numeric(scale(surveys$julian.date))
surveys$n_bees_scaled <- as.numeric(scale(surveys$n_bees))
```

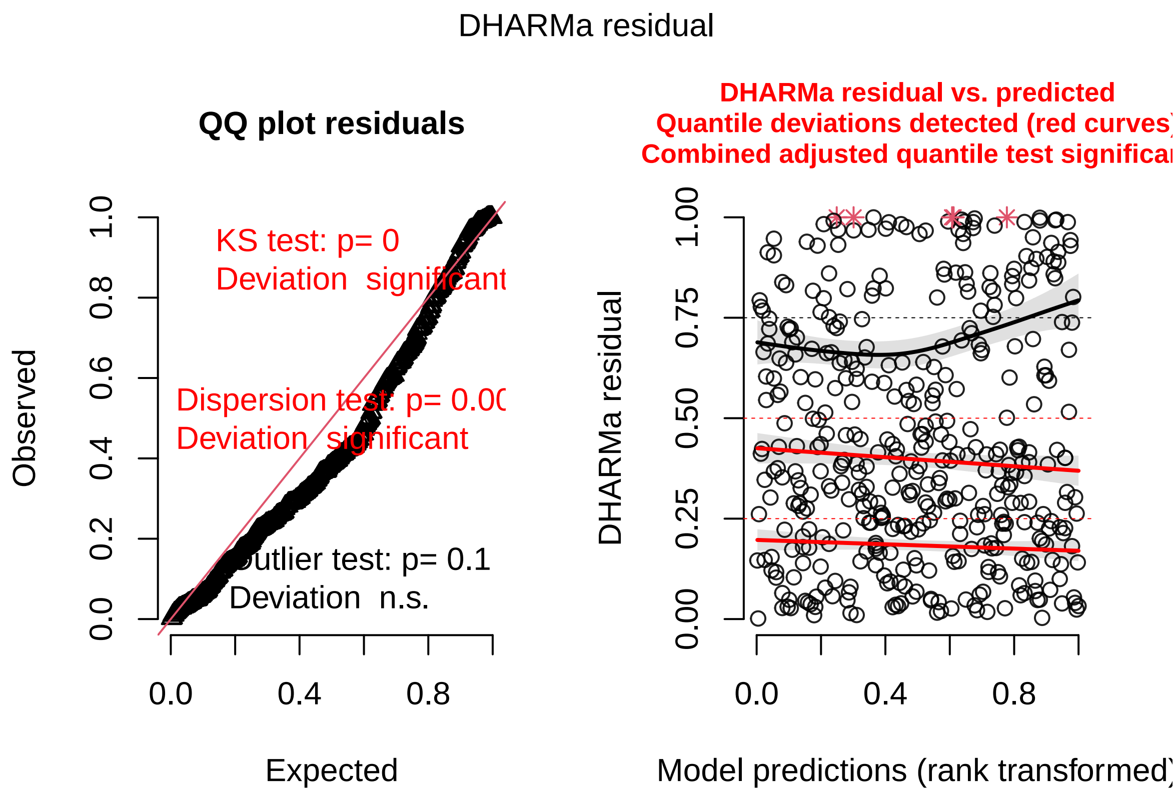
2. Run poisson model

```
caste_glm_full <- glmmTMB(n_bees ~ caste*julian.date_scaled*plot_type +
                            cover_type +
                              (1|area/site_id/plot_id),
                            family = poisson,
                            data = surveys)
```

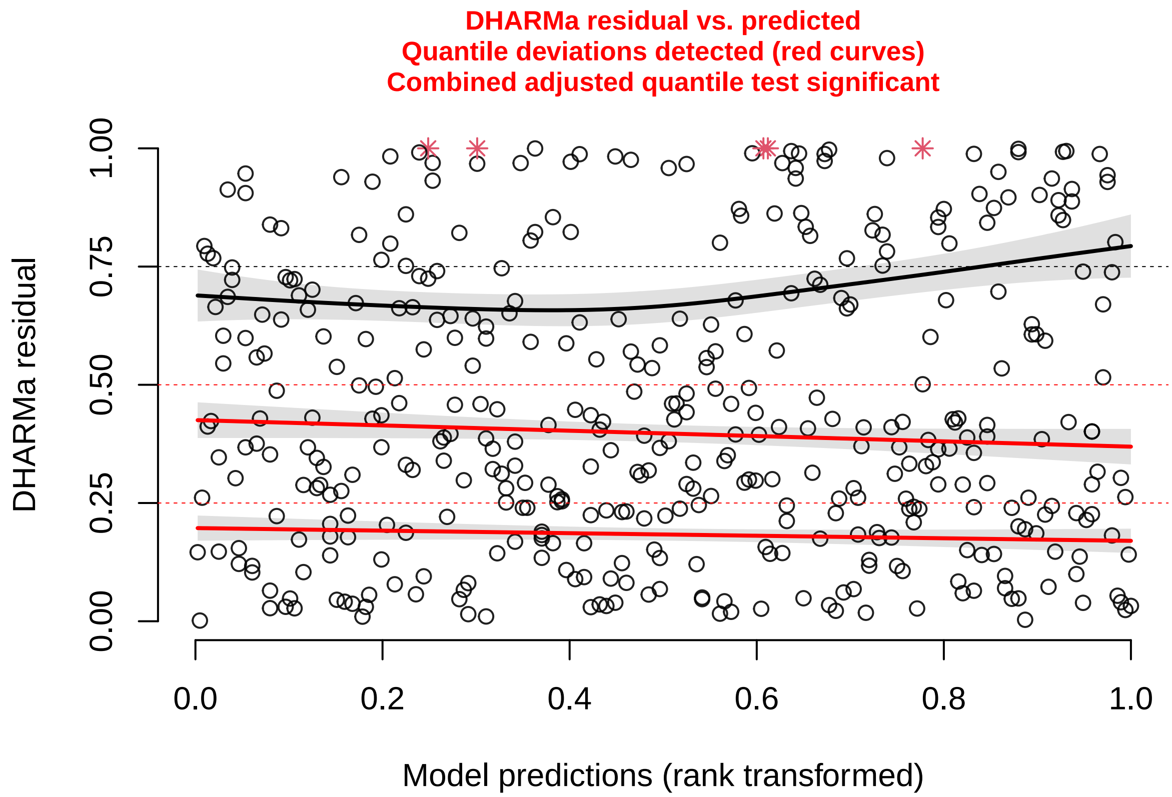
3. Check diagnostics of model

- significant deviations

```
#simulate residuals
residuals_caste_glm_full <- simulateResiduals(fittedModel = caste_glm_full, plot = TRUE)
```

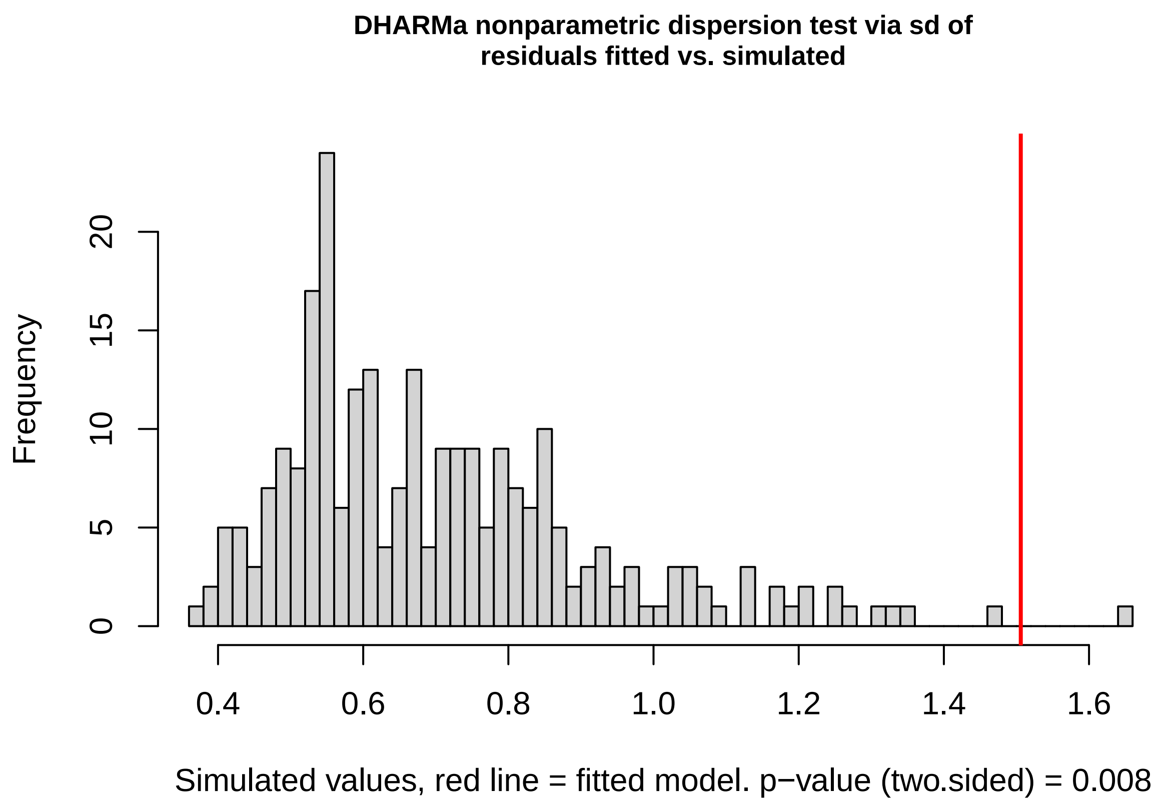



```
testQuantiles(caste_glm_full) #significant
```



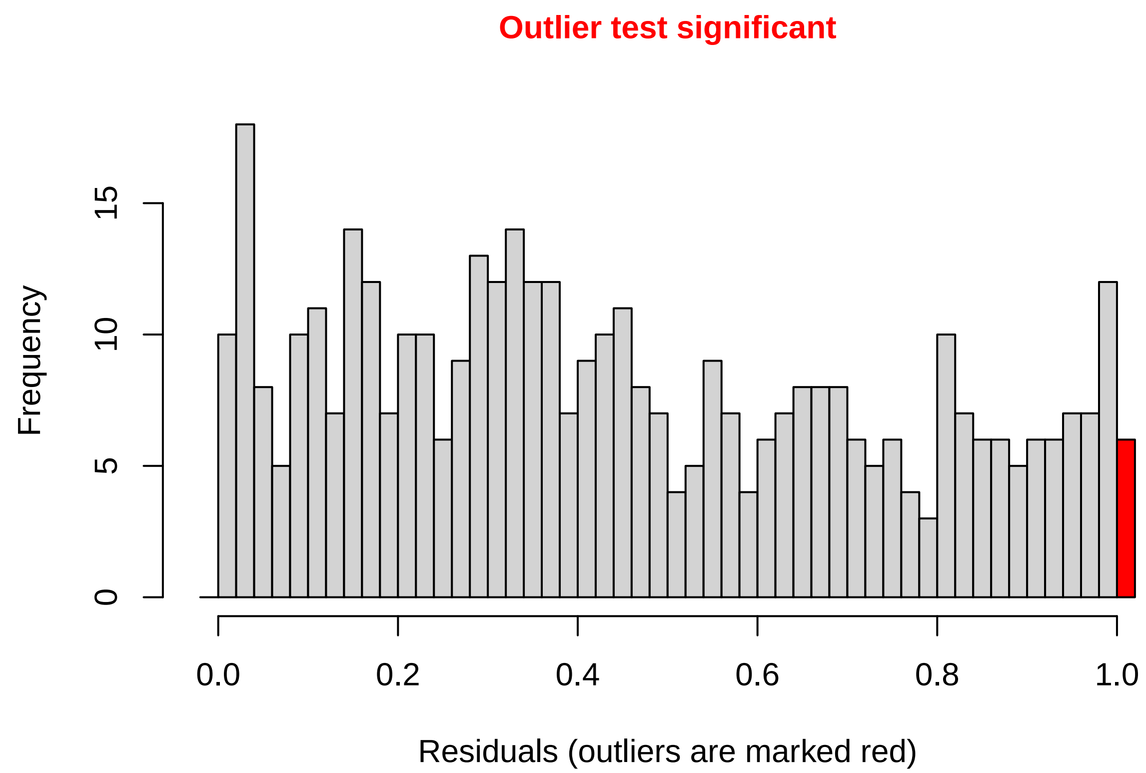

```
##
##  Test for location of quantiles via qgam
##
## data:  res
## p-value = 9.523e-08
## alternative hypothesis: both
```

```
testDispersion(caste_glm_full) #significant
```



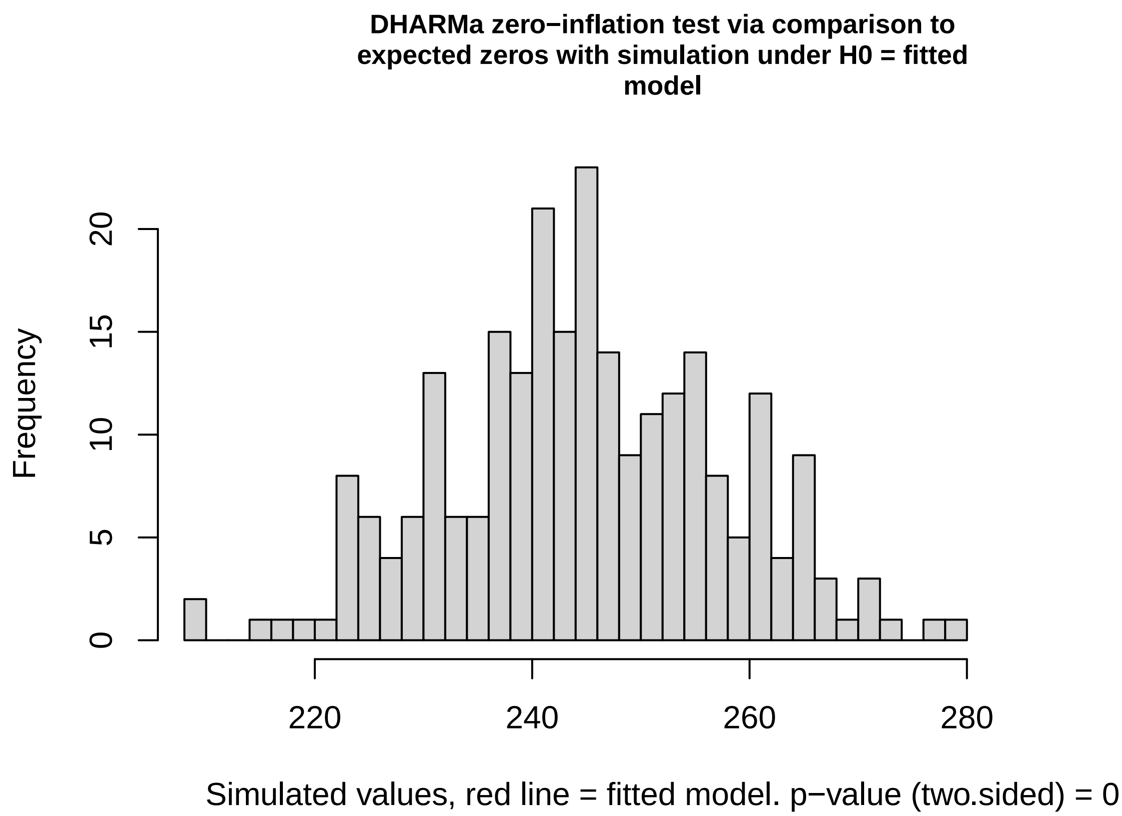

```
##
##  DHARMa nonparametric dispersion test via sd of residuals fitted vs.
##  simulated
##
## data:  simulationOutput
## dispersion = 2.1315, p-value = 0.008
## alternative hypothesis: two.sided
```

```
testOutliers(caste_glm_full) #significant
```




```
##
##  DHARMa bootstrapped outlier test
##
## data:  caste_glm_full
## outliers at both margin(s) = 6, observations = 420, p-value = 0.02
## alternative hypothesis: two.sided
##  percent confidence interval:
##  0.00000000 0.01077381
## sample estimates:
## outlier frequency (expected: 0.00330952380952381 )
##                                          0.01428571
```

```
testZeroInflation(caste_glm_full) #significant
```



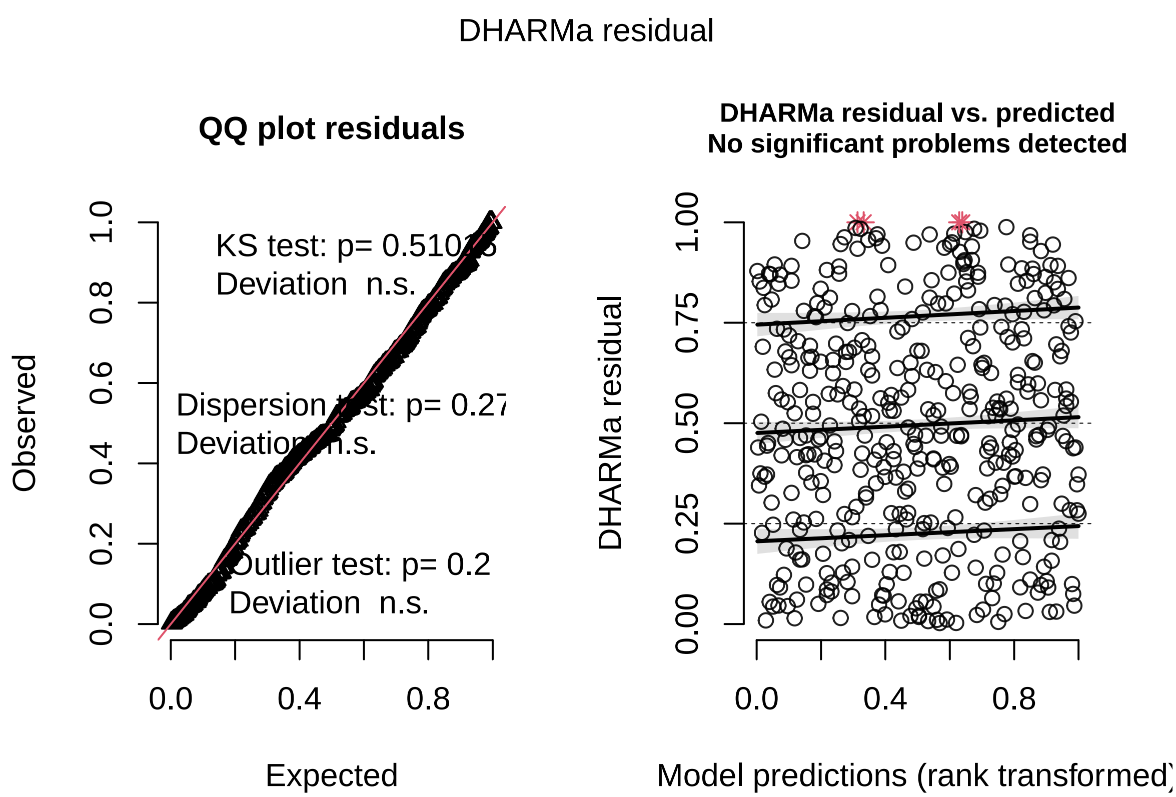

```
##
##  DHARMa zero-inflation test via comparison to expected zeros with
##  simulation under H0 = fitted model
##
## data:  simulationOutput
## ratioObsSim = 1.2023, p-value < 2.2e-16
## alternative hypothesis: two.sided
```

4. Try negative binomial

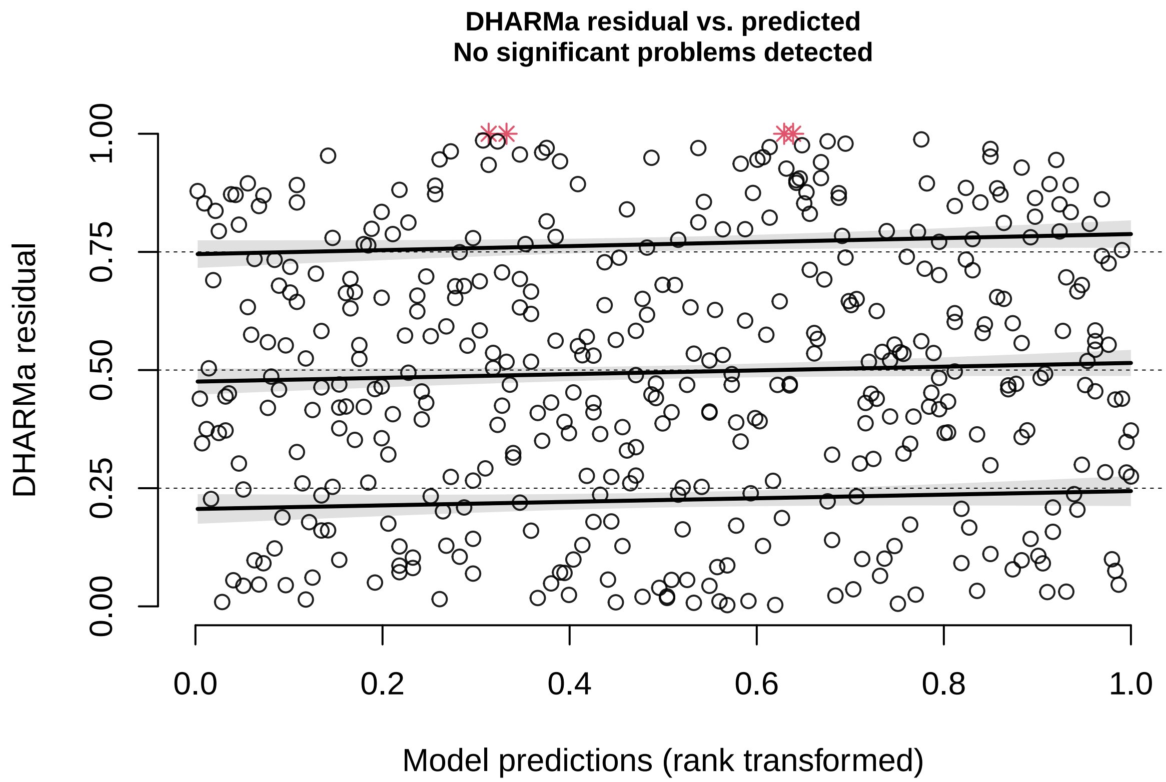
- looks good

```
caste_glm_nbin <- glmmTMB(n_bees ~ caste*julian.date_scaled*plot_type +
                            cover_type +
                              (1|area/site_id/plot_id),
                            family = nbinom2(),
                            data = surveys)

#simulate residuals
residuals_caste_glm_nbin <- simulateResiduals(fittedModel = caste_glm_nbin, plot = TRUE)
```

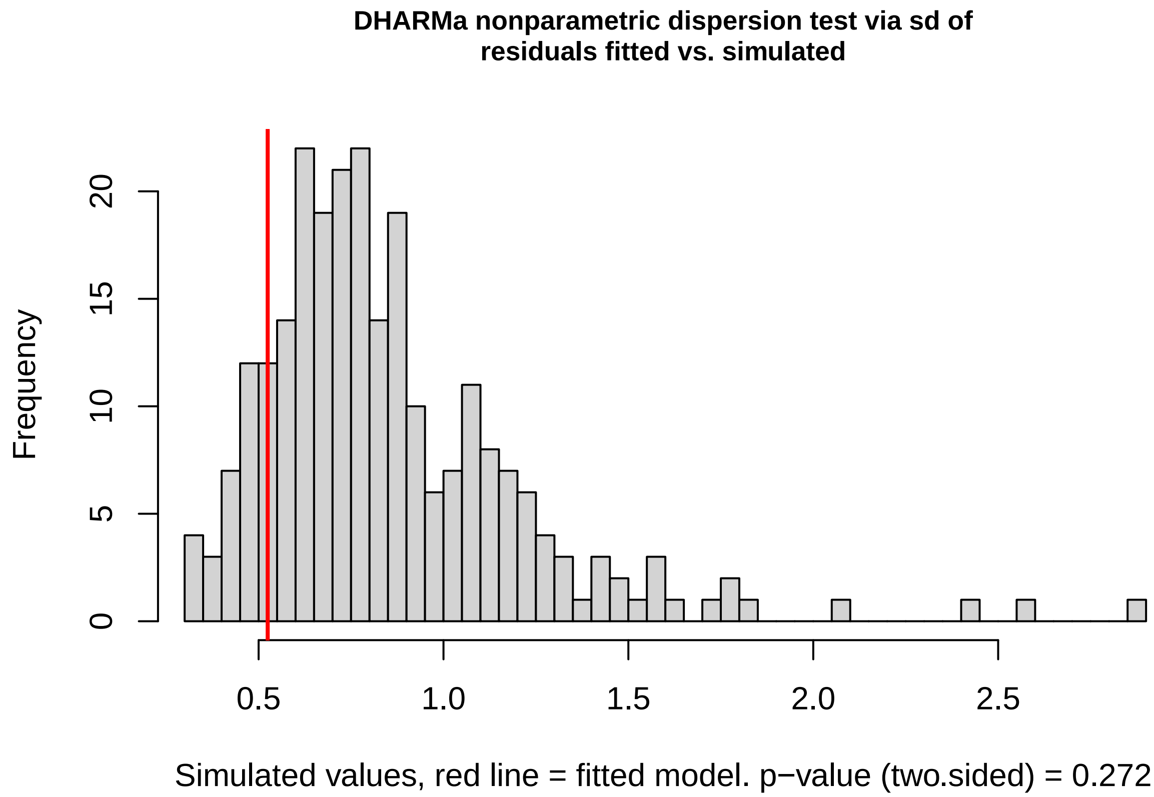



```
testQuantiles(caste_glm_nbin) #ns
```



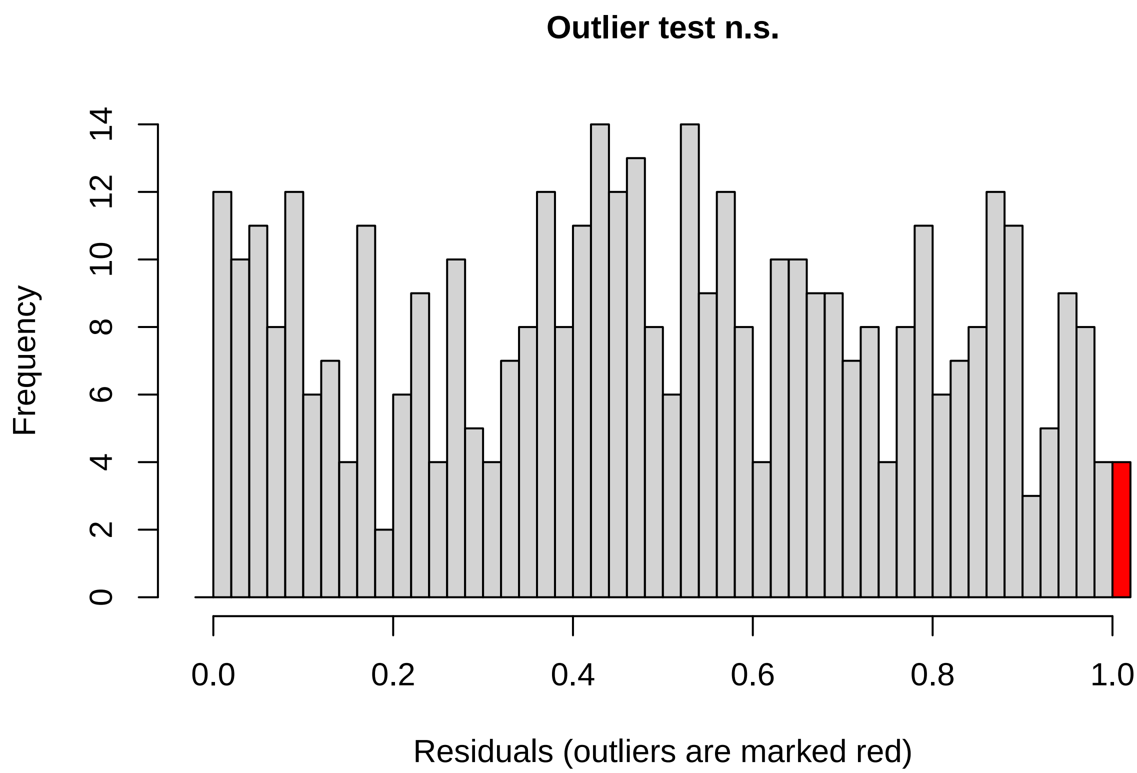

```
##
##  Test for location of quantiles via qgam
##
## data:  res
## p-value = 0.598
## alternative hypothesis: both
```

```
testDispersion(caste_glm_nbin) #ns
```



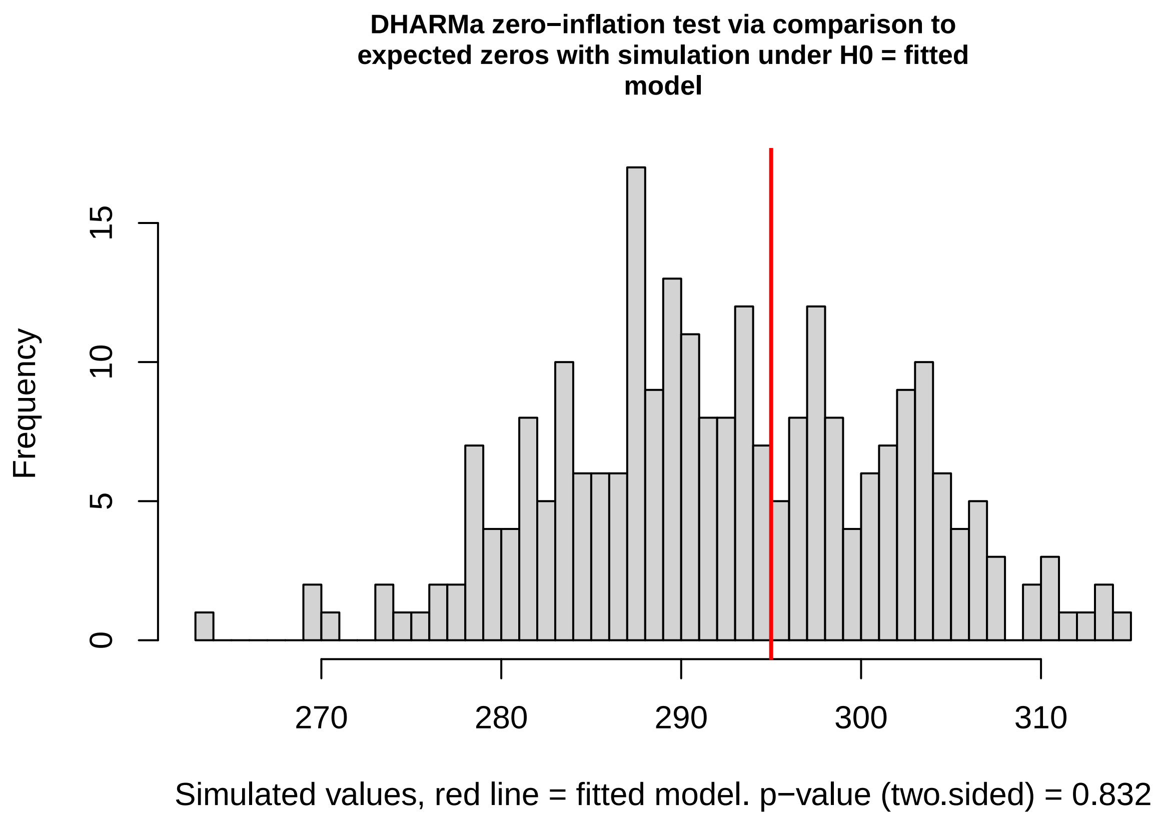

```
##
##  DHARMa nonparametric dispersion test via sd of residuals fitted vs.
##  simulated
##
## data:  simulationOutput
## dispersion = 0.61325, p-value = 0.272
## alternative hypothesis: two.sided
```

```
testOutliers(caste_glm_nbin) #ns
```




```
##
##  DHARMa bootstrapped outlier test
##
## data:  caste_glm_nbin
## outliers at both margin(s) = 4, observations = 420, p-value = 0.12
## alternative hypothesis: two.sided
##  percent confidence interval:
##  0.00000000 0.00952381
## sample estimates:
## outlier frequency (expected: 0.00414285714285714 )
##                                           0.00952381
```

```
testZeroInflation(caste_glm_nbin) #ns
```




```
##
##  DHARMa zero-inflation test via comparison to expected zeros with
##  simulation under H0 = fitted model
##
## data:  simulationOutput
## ratioObsSim = 1.0072, p-value = 0.832
## alternative hypothesis: two.sided
```

5. Plot results

```
summary(caste_glm_nbin)
```

```
##  Family: nbinom2  ( log )
## Formula:
## n_bees ~ caste * julian.date_scaled * plot_type + cover_type +
##     (1 | area/site_id/plot_id)
## Data: surveys
##
##       AIC       BIC    logLik -2*log(L)  df.resid
##     897.4     949.9    -435.7     871.4       407
##
## Random effects:
##
## Conditional model:
##  Groups               Name        Variance  Std.Dev.
```

```
##  plot_id:site_id:area (Intercept) 2.244e-02 1.498e-01
##  site_id:area         (Intercept) 6.104e-02 2.471e-01
##  area                 (Intercept) 2.800e-09 5.291e-05
## Number of obs: 420, groups:
## plot_id:site_id:area, 36; site_id:area, 12; area, 6
##
## Dispersion parameter for nbinom2 family (): 0.451
##
## Conditional model:
##                                            Estimate Std. Error z value Pr(>|z|)
## (Intercept)                                -0.01592    0.28369  -0.056  0.95524
## castew                                     -1.18848    0.39976  -2.973  0.00295
## julian.date_scaled                         -0.66541    0.23278  -2.859  0.00426
## plot_typefield                             -0.19879    0.28410  -0.700  0.48409
## cover_typegrass                             0.13793    0.25977   0.531  0.59545
## castew:julian.date_scaled                   2.00065    0.46463   4.306 1.66e-05
## castew:plot_typefield                      -0.15039    0.48337  -0.311  0.75570
## julian.date_scaled:plot_typefield          -0.24601    0.30398  -0.809  0.41834
## castew:julian.date_scaled:plot_typefield   -0.65550    0.56130  -1.168  0.24287
##
## (Intercept)
## castew                                     **
## julian.date_scaled                         **
## plot_typefield
## cover_typegrass
## castew:julian.date_scaled                  ***
## castew:plot_typefield
## julian.date_scaled:plot_typefield
## castew:julian.date_scaled:plot_typefield
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Plots

```r
#set peak bloom start and end dates
bloom_start <- as.Date("2024-04-19")
bloom_end <- as.Date("2024-05-24")

## Create date column
surveys <- surveys %>%
  mutate(
    # Convert Day of Year to an actual Date object (using 2024 as a reference year)
    date_formatted = as.Date(round(julian.date) - 1, origin = "2024-01-01")
  )

ggplot(surveys, aes(x = date_formatted, y = n_bees, color = plot_type)) +
  # Add the bloom window as a shaded background
  annotate("rect", xmin = bloom_start, xmax = bloom_end, ymin = -Inf, ymax = Inf,
           alpha = .2, fill = "steelblue") +
  geom_jitter(
    width = 1,
```

```
    height = 0,
    size = 2.5,
    alpha = 0.2
  ) +
  stat_summary(fun = mean, geom = "line", linewidth = 1.2) +
  facet_wrap(~ caste, ncol = 1, scales = "free_y", labeller = as_labeller(c("w" = "Workers", "q" = "Quee
  labs(x = "Date", y = "Bumble bee count\n(per plot)", color = "Plot type", fill = "Plot type") +
  theme_classic(base_size = 20)
```

Plot behaviours of each caste in each plot type

```
#add plot type column to bee dataframe
bee <- bee %>%
  mutate(plot_type = if_else(plot_id == "d", "ditch", "field"))
bee <- bee %>%
   mutate(
     plot_type = as.factor(plot_type))
# Summary data for behavior proportions
behavior_summary <- bee %>%
  filter(caste %in% c("q", "w")) %>%
  group_by(plot_type, caste, behaviour) %>%
  summarise(total_count = n(), .groups = "drop") %>%
  group_by(plot_type, caste) %>%
```

```
  mutate(proportion = total_count / sum(total_count))

# Plotting the breakdown
ggplot(behavior_summary, aes(x = plot_type, y = proportion, fill = behaviour)) +
  geom_bar(stat = "identity", position = "fill") +
  facet_wrap(~caste, labeller = as_labeller(c("w" = "Workers", "q" = "Queens"))) +
  scale_y_continuous(labels = scales::percent) +
  scale_fill_discrete(labels = c(
    "flying" = "Flying",
    "foraging" = "Foraging",
    "nest_searching" = "Nest searching",
    "resting" = "Resting"
  )) +
  labs(
    x = "Plot type",
    y = "Percentage of observations",
    fill = "Behaviour"
  ) +
  theme_classic(base_size = 15)
```

Number of individuals observed

```
#number of individuals seen in each caste
queen_num <- count(bee, caste == "q") ##262
worker_num <- count(bee, caste == "w") ##77
male_num <- count(bee, caste == "m") ##39
unknown_num <- count(bee, caste == "") ##142
total_bee <- count(bee) ##321
##put into table
bee_count_summary <- bee %>%
  #change blanks to unknown
  mutate(caste = ifelse(caste == "" , "unknown", caste)) %>%
  # Group by species and caste
  group_by(caste) %>%
  summarise(n = n(), .groups = "drop") %>%
  # Add a total row
  bind_rows(tibble(caste = "overall", n = sum(.$n)))

#number of nest-searching queens
nsq_num <- count(bee, caste == "q", behaviour == "nest_searching") ##93
```

Calculate percentage of visits to blueberry for each caste of Bombus

```
#number of individuals visiting VACO in each caste
queen_vaco_num <- count(bee, caste == "q", flower_code == "vaco") ##64
worker_vaco_num <- count(bee, caste == "w", flower_code == "vaco") ##27
male_vaco_num <- count(bee, caste == "m", flower_code == "vaco") ##1
unknown_vaco_num <- count(bee, caste == "", flower_code == "vaco") ##25

#total vaco visits by bombus, not including unknowns
vaco_visit <- count(bee, flower_code == "vaco", caste != "") ##92

#percentage of total VACO visit
queen_vaco_num/vaco_visit #0.70
```

```
##   caste == "q" flower_code == "vaco"         n
## 1          NaN                   NaN 1.7521368
## 2          NaN                     1 0.1853147
## 3            1                   NaN 7.9200000
## 4            1                     1 0.6956522
```

```
worker_vaco_num/vaco_visit #0.29
```

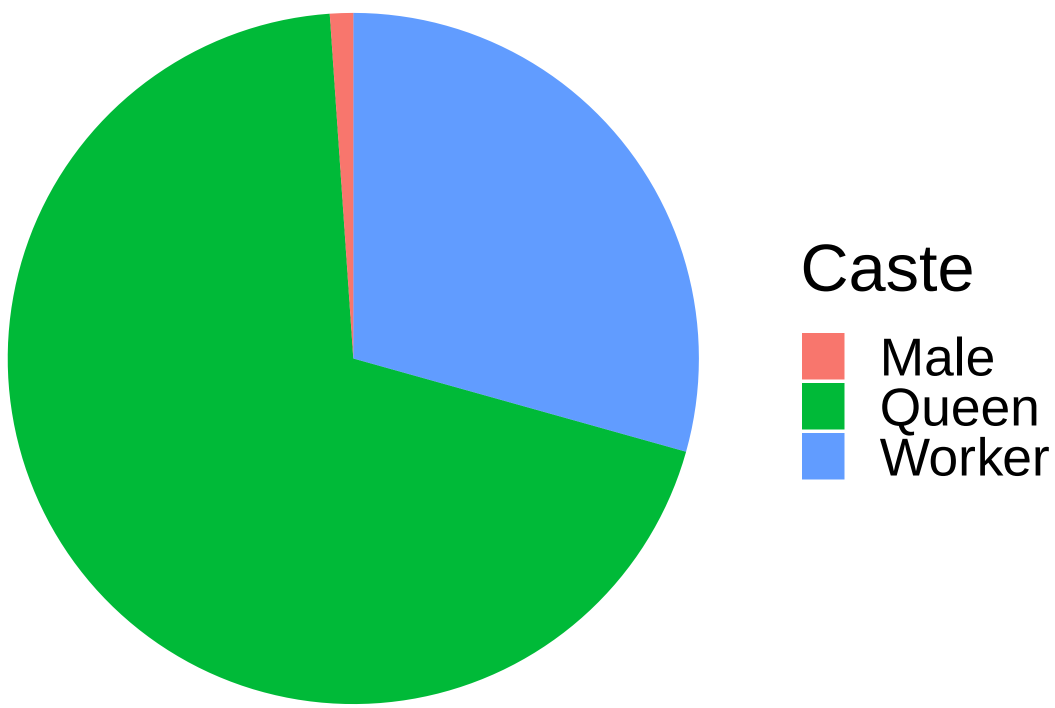
```
##   caste == "w" flower_code == "vaco"         n
## 1          NaN                   NaN 3.0170940
## 2          NaN                     1 0.3146853
## 3            1                   NaN 2.0000000
## 4            1                     1 0.2934783
```

```r
male_vaco_num/vaco_visit #0.01
```

```
##   caste == "m" flower_code == "vaco"          n
## 1          NaN                  NaN 3.11965812
## 2          NaN                    1 0.40559441
## 3            1                  NaN 1.52000000
## 4            1                    1 0.01086957
```

```r
vaco_caste <- data.frame(
  caste = c("Queen", "Worker", "Male"),
  count = c(64, 27, 1)
)

ggplot(vaco_caste, aes(x = "", y = count, fill = caste)) +
  geom_col(width = 1) +
  coord_polar(theta = "y") +
  theme_void(base_size = 25) +
  labs(fill = "Caste")
```

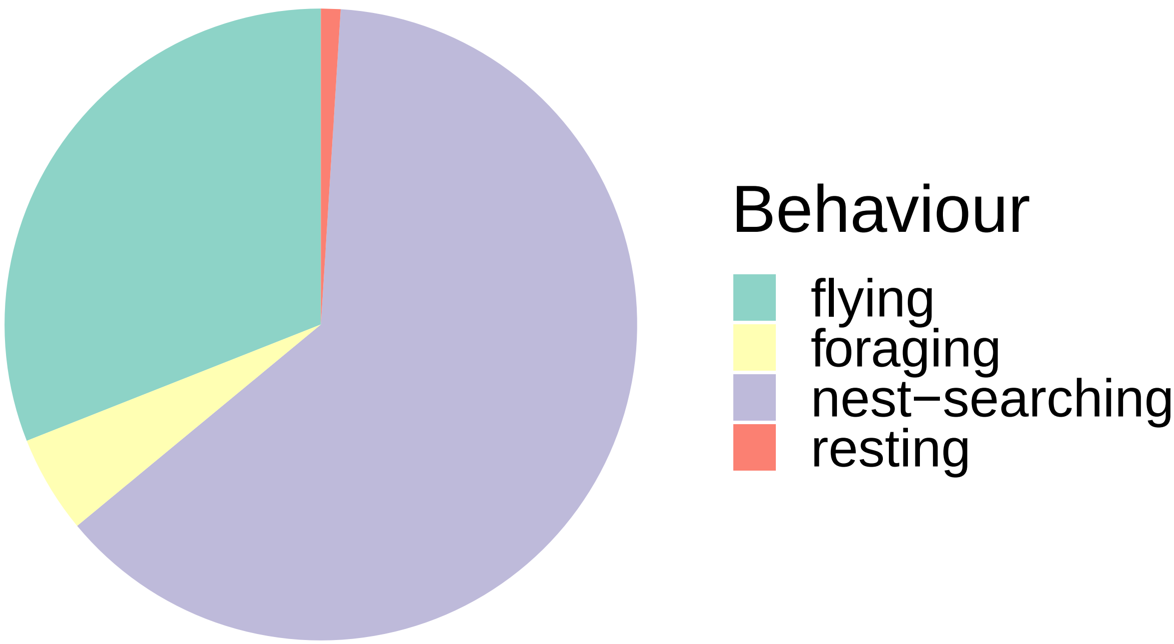


Calculate percentage of observations for each behaviour in each plot type for each caste

```r
#queen ditch
behaviour_queen_ditch <- data.frame(
  caste = c("nest-searching", "flying", "foraging", "resting"),
  count = c(63, 31, 5,1)
```

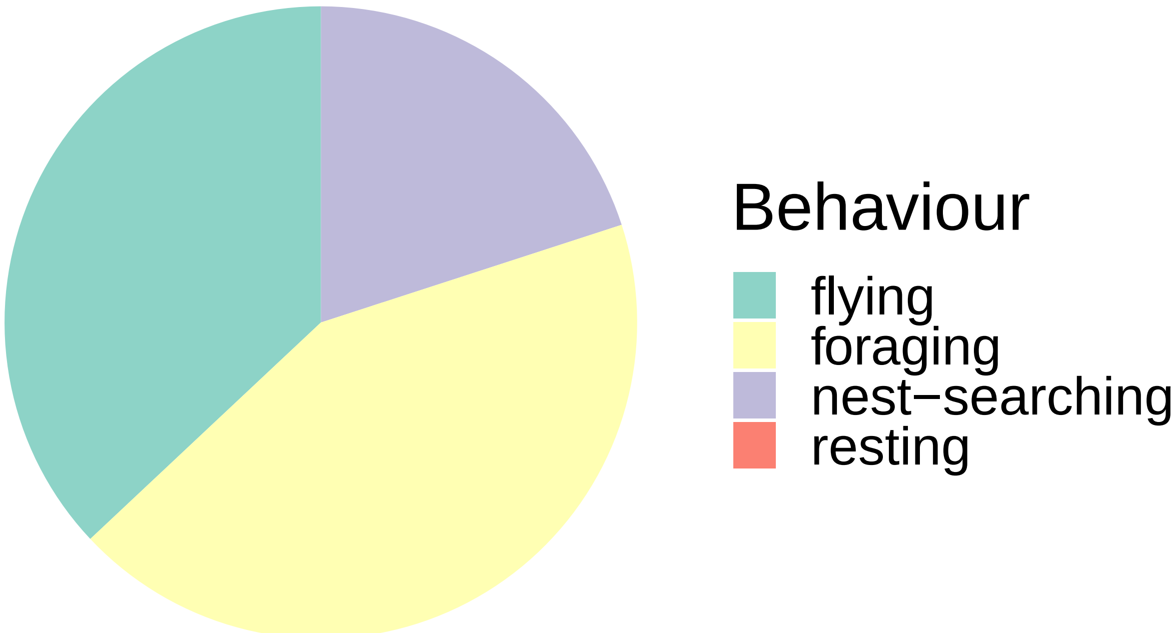
```
)

ggplot(behaviour_queen_ditch, aes(x = "", y = count, fill = caste)) +
  geom_col(width = 1) +
  coord_polar(theta = "y") +
  theme_void(base_size = 25) +
  scale_fill_brewer(palette = "Set3") +
  labs(fill = "Behaviour")
```



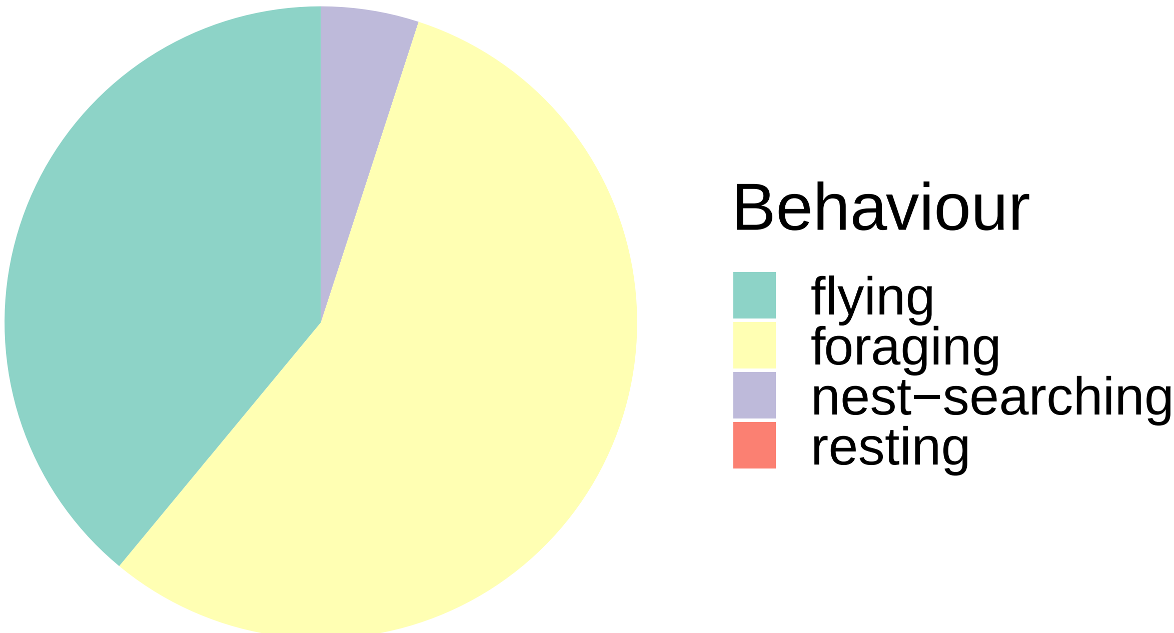

```
#queen field
behaviour_queen_field <- data.frame(
  caste = c("nest-searching", "flying", "foraging", "resting"),
  count = c(20, 37, 43,0)
)

ggplot(behaviour_queen_field, aes(x = "", y = count, fill = caste)) +
  geom_col(width = 1) +
  coord_polar(theta = "y") +
  theme_void(base_size = 25) +
  scale_fill_brewer(palette = "Set3") +
  labs(fill = "Behaviour")
```



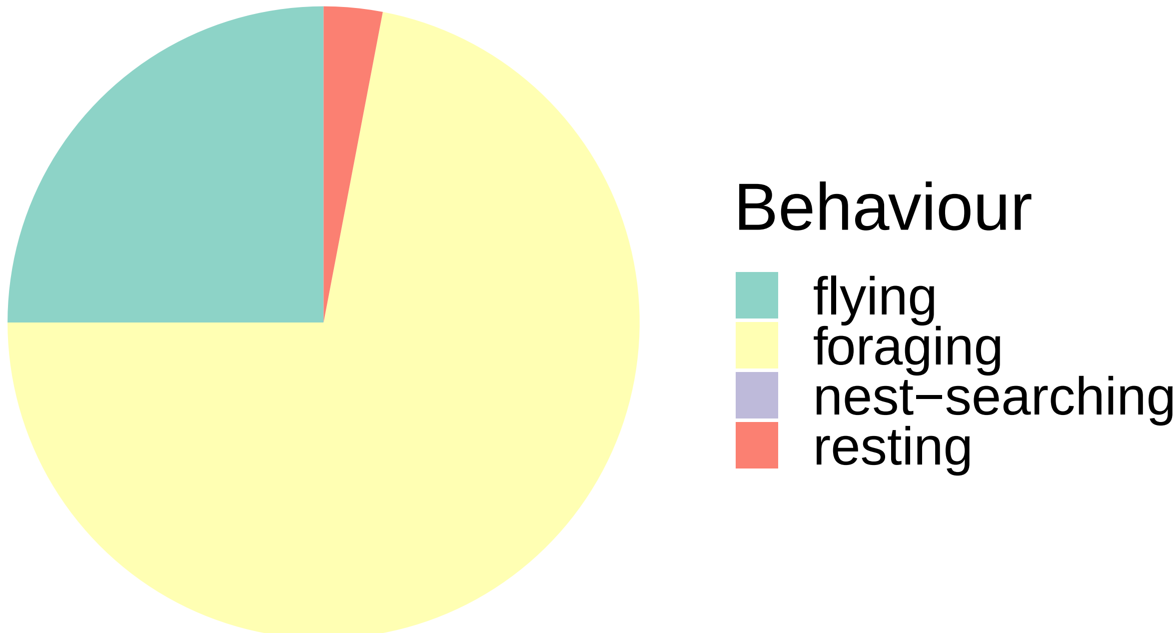

```r
#worker ditch
behaviour_worker_ditch <- data.frame(
  caste = c("nest-searching", "flying", "foraging", "resting"),
  count = c(5, 39, 56,0)
)

ggplot(behaviour_worker_ditch, aes(x = "", y = count, fill = caste)) +
  geom_col(width = 1) +
  coord_polar(theta = "y") +
  theme_void(base_size = 25) +
  scale_fill_brewer(palette = "Set3") +
  labs(fill = "Behaviour")
```




```r
#worker field
behaviour_worker_field <- data.frame(
  caste = c("nest-searching", "flying", "foraging", "resting"),
  count = c(0, 25, 72, 3)
)

ggplot(behaviour_worker_field, aes(x = "", y = count, fill = caste)) +
  geom_col(width = 1) +
  coord_polar(theta = "y") +
  theme_void(base_size = 25) +
  scale_fill_brewer(palette = "Set3") +
  labs(fill = "Behaviour")
```


Behaviour
flying
foraging
nest−searching
resting